Computer Architecture – Full Syllabus

Module 1: Introduction to Computer Architecture

- Difference between Computer Architecture and Computer Organization
- Basic functional units of a computer
- Von Neumann Architecture
- Harvard Architecture
- System Bus and Bus Structure
- Instruction Execution Cycle
- Performance Metrics: Latency, Throughput, MIPS, FLOPS, CPI

Module 2: Data Representation and Number Systems

- Binary, Octal, Decimal, Hexadecimal number systems
- Signed and unsigned numbers
- 1's and 2's complement representation
- Binary Arithmetic: Addition, Subtraction, Multiplication, Division
- Floating Point Representation (IEEE 754 Standard)
- Character Representation: ASCII and Unicode

Module 3: Processor and Instruction Set Architecture (ISA)

- Types of instructions: Data movement, Arithmetic, Logical, Control
- Instruction Formats: R-type, I-type, J-type
- Addressing Modes: Immediate, Direct, Indirect, Register, Indexed, Base + Offset
- Instruction Cycle: Fetch, Decode, Execute
- RISC vs CISC Architecture
- Stack vs Accumulator vs Register-based ISA

Module 4: Central Processing Unit (CPU)

- CPU Components: ALU, Control Unit, Registers
- Register Organization
- Instruction Cycle (Detailed)
- Hardwired vs Microprogrammed Control
- Control Signal Generation
- CPU Performance Improvement Techniques: Pipelining, Superscalar

Module 5: Pipelining and Hazards

- Basic Concepts of Pipelining
- Pipeline Stages: Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), Write Back (WB)

- Pipeline Hazards:
    - Structural Hazard
    - Data Hazard (RAW, WAR, WAW)
    - Control Hazard (Branch Hazards)
- Techniques to Handle Hazards: Forwarding, Stalling, Branch Prediction

Module 6: Memory Organization

- Types of Memory: RAM, ROM, SRAM, DRAM
- Memory Hierarchy
- Main Memory vs Cache vs Virtual Memory
- Cache Mapping Techniques: Direct Mapping, Associative Mapping, Set-Associative Mapping
- Cache Replacement Policies: LRU, FIFO, Random
- Memory Interleaving
- Paging, Segmentation, Translation Lookaside Buffer (TLB)

Module 7: Input/Output (I/O) Organization

- I/O Devices and Interfaces
- Programmed I/O
- Interrupt-Driven I/O
- Direct Memory Access (DMA)
- I/O Mapped I/O vs Memory Mapped I/O
- Priority Interrupts
- I/O Buses

Module 8: Arithmetic Logic Unit (ALU) & Digital Circuits

- ALU Design
- Combinational Logic Circuits: Adders, Subtractors, Multipliers
- Sequential Circuits: Flip-Flops, Counters, Registers
- Booth's Algorithm for Multiplication
- Restoring and Non-Restoring Division Algorithms
- Floating Point Arithmetic

Module 9: Control Unit Design

- Functions of the Control Unit
- Micro-operations
- Hardwired Control
- Microprogrammed Control
- Control Word and Control Memory

Module 10: Parallel and Advanced Architectures

- Flynn’s Classification: SISD, SIMD, MISD, MIMD
- Multi-core Processors
- Superscalar Architecture
- Multiprocessing
- Interconnection Networks: Crossbar, Bus, Mesh
- Shared vs Distributed Memory Systems

Optional (Advanced Topics)

- Instruction Level Parallelism (ILP)
- Very Long Instruction Word (VLIW)
- Pipeline Scheduling
- Memory Consistency Models
- GPU Architecture Basics